

Safe Infrastructure or Risk Signal? Predicting Serious Collision Severity around Pedestrian Crossing Facilities in London

Preparation

- [Github link](#)
- Number of words: 1550
- Runtime: 2min (Tested in VS Code using base (Python 3.13.5)) Chip: Apple M1 Memory: 8 GB
- Coding environment: VS Code Jupyter Notebook, Python 3.13.5, conda base environment
- Execution note: If running via SDS Docker or other Jupyter environments, the kernel may crash ("Kernel Died") due to memory constraints. The notebook was therefore developed and successfully executed in VS Code using Python.
- License: this notebook is made available under the [Creative Commons Attribution license](#).
- Additional library [*libraries not included in SDS Docker or not used in this module*]:
 - **SHAP**: used for model interpretability.
 - **Statsmodels**: used for VIF diagnostics.
- AI tools were used for code debugging and wording refinement. The final work remains the author's own original work.

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Introduction

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In London, Vision Zero aims to eliminate deaths and serious injuries from the transport network, making understanding of collision severity important for road safety planning (Mayor of London, n.d.).

Existing research has used machine learning to predict pedestrian crash severity at signalised intersections, showing that spatial and traffic-related factors support severity prediction (Kuşkan et al., 2022). Built environment studies have used XGBoost and SHAP to explain fatal pedestrian accident risk at specific locations (Chang et al., 2022). Recent intersection research extends the 5D built-environment framework with detailed intersection-interface features (Chen et al., 2026). Crossing safety studies show that facilities are not always protective: collision risk can be higher at some signalised crossings, and unsignalised zebra crossings can involve fatality risk (Quistberg et al., 2015; Olszewski et al., 2015).

Crossing facilities are often treated as protective infrastructure, but they may also mark high-conflict locations with complex junction layouts. Although much of the crossing-facility literature focuses on pedestrian safety, this study examines all reported collisions, treating crossing facilities as observable road-environment features. Their role has not been tested within an interpretable machine-learning prediction framework using London-wide collision data.

Research questions

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This study addresses two research questions.

RQ1: How do severe collision rates vary across pedestrian crossing facility types and junction contexts in London in 2024?

RQ2: Can machine learning models identify serious or fatal collisions in London, and which road-context variables contribute most to severity prediction?

Data

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```
In [1]: # 3.1 Install and import
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import requests

from statsmodels.stats.outliers_influence import variance_inflation
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler, FunctionTransformer
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.pipeline import Pipeline
from sklearn.metrics import matthews_corrcoef
from sklearn.metrics import (
    classification_report, confusion_matrix,
    ConfusionMatrixDisplay, precision_score,
    recall_score, f1_score, accuracy_score,
    roc_auc_score, average_precision_score,
    precision_recall_curve
)

warnings.filterwarnings('ignore')
print("All packages imported successfully")
```

All packages imported successfully

Data sources and preparation

This study uses the STATS19 collision dataset for 2024, published by the Department for Transport (2024). The GB dataset contains 100,927 records, filtered to Greater London using ONS codes beginning with E09, retaining 20,967 collisions across 33 boroughs. The collision_index field was used solely for merging and excluded from model features. The casualty dataset was not used as a model input. The analysis focuses on road-context variables at the collision level.

```
In [2]: # 3.2 Load collision and casualty datasets from GitHub

COLLISION_URL = (
    'https://raw.githubusercontent.com/Alexia418/'
    'DSSS_Assessment/main/'
    'dft-road-casualty-statistics-collision-2024.csv'
)

CASUALTY_URL = (
    'https://raw.githubusercontent.com/Alexia418/'
    'DSSS_Assessment/main/'
```

```

        'dft-road-casualty-statistics-casualty-2024.csv'
    )

    collision = pd.read_csv(COLLISION_URL, low_memory=False)
    casualty = pd.read_csv(CASUALTY_URL, low_memory=False)

    print(f"Collision records (GB): {len(collision):,}")
    print(f"Casualty records (GB): {len(casualty):,}")

```

Collision records (GB): 100,927
 Casualty records (GB): 128,272

In [3]: *# 3.3 Filter to London using ONS district codes (E09 = London boroughs)*

```

london_col = collision[
    collision['local_authority_ons_district']
        .astype(str).str.startswith('E09')
].copy()

london_cas = casualty[
    casualty['collision_index'].isin(london_col['collision_index'])
].copy()

print(f"London collisions: {len(london_col):,}")
print(f"London casualties: {len(london_cas):,}")
print(f"Unique boroughs: {len(london_col['local_authority_ons_district']):,}")

```

London collisions: 20,967
 London casualties: 23,997
 Unique boroughs: 33

In [4]: *# 3.4 Aggregate pedestrian casualties to collision level and join
 # casualty_class = 3: Pedestrian; cross-validated against casualty_severity*

```

ped = london_cas[london_cas['casualty_class'] == 3].copy()

ped_agg = ped.groupby('collision_index').agg(
    ped_casualty_count=('casualty_severity', 'count'),
    ped_severe_count=('casualty_severity', lambda x: (x.isin([1,2])).sum()),
    has_ped_casualty=('casualty_severity', lambda x: 1 if x.any() else 0)
).reset_index()

london = london_col.merge(ped_agg, on='collision_index', how='left')
london['has_ped_casualty'] = london['has_ped_casualty'].fillna(0).astype(int)

print(f"Pedestrian casualties: {len(ped):,}")
print(f"Merged dataset rows: {len(london):,}")

```

Pedestrian casualties: 4,445
 Merged dataset rows: 20,967

In [5]: *# 3.5 Create binary target variable
 # collision_severity: 1=Fatal, 2=Serious, 3=Slight (STATS19)*

```

london['severe'] = london['collision_severity'].apply(
    lambda x: 1 if x in [1, 2] else 0
)

```

```
print(f"Severe cases: {london['severe'].sum():,} ({london['severe'].sum()/london['severe'].count():.1%})")
print(f"Slight cases: {(london['severe']!=0).sum():,} ({(london['severe']!=0).sum()/london['severe'].count():.1%})")
```

Severe cases: 3,601 (17.2%)

Slight cases: 17,366

Feature engineering and cleaning

The binary target was derived from collision_severity: severe (1) for fatal or serious collisions (codes 1 and 2), slight (0) for non-serious (code 3). Two temporal features were derived: hour of day and binary indicators for peak hour (07:00–09:00 and 16:00–19:00) and weekend. Rows with ambiguous or unknown codes were removed using variable-specific rules. The largest removals came from pedestrian crossing facility type (3,690 rows) and junction control (3,718 rows). After all exclusions, the final dataset contains 11,762 collisions, of which 2,266 (19.3%) are severe (4.2:1 imbalance). No missing values remained. All categorical variables were one-hot encoded using pd.get_dummies (53 features total). Numeric features (speed_limit, hour) were retained in continuous form.

In [6]: *# 3.6 Engineer temporal and spatial features*

```
london['hour'] = london['time'].str[:2].astype(float)
london['peak_hour'] = london['hour'].apply(lambda x: 1 if (7<=x<=19) else 0)
london['weekend'] = london['day_of_week'].apply(lambda x: 1 if x in [6, 7] else 0)

print("Features created: hour, peak_hour, weekend")
```

Features created: hour, peak_hour, weekend

In [7]: *# 3.8 Inspect ambiguous and missing codes before exclusion*

```
pd.set_option('display.max_rows', 20)
pd.set_option('display.width', 120)

AMBIGUOUS_CODES = {
    'pedestrian_crossing_physical_facilities_historic': [-1, 9],
    'junction_detail': [-1, 99],
    'junction_control': [-1, 9],
    'road_type': [-1, 9],
    'speed_limit': [-1, 99],
    'light_conditions': [-1],
    'weather_conditions': [-1, 9],
    'road_surface_conditions': [-1, 9],
    'urban_or_rural_area': [-1],
    'day_of_week': [-1],
}

rows = []
for col, codes in AMBIGUOUS_CODES.items():
    if col not in london.columns:
        continue
    total = len(london)
```

```

ambig      = london[col].isin(codes).sum()
nan        = london[col].isnull().sum()
total_miss = ambig + nan
rows.append({'variable': col, 'ambiguous': ambig,
            'NaN': nan, 'total_missing': total_miss,
            'pct': round(total_miss/total*100, 1)})

print(pd.DataFrame(rows).sort_values('pct', ascending=False).to_str

```

	variable	ambiguous	NaN	to
tal_missing	pct			
	junction_control	5880	0	
5880	28.0			
	pedestrian_crossing_physical_facilities_historic	3690	0	
3690	17.6			
	junction_detail	2256	0	
2256	10.8			
	road_type	2216	0	
2216	10.6			
	weather_conditions	1905	0	
1905	9.1			
	road_surface_conditions	1625	0	
1625	7.8			
0	0.0	0	0	
	light_conditions	0	0	
0	0.0			
	urban_or_rural_area	0	0	
0	0.0			
	day_of_week	0	0	
0	0.0			

In [8]: # 3.9 Define model features and target

```

FEATURES = [
    'pedestrian_crossing_physical_facilities_historic',
    'junction_detail',
    'junction_control',
    'road_type',
    'speed_limit',
    'urban_or_rural_area',
    'light_conditions',
    'weather_conditions',
    'road_surface_conditions',
    'day_of_week',
    'hour',
    'peak_hour',
    'weekend',
]

TARGET = 'severe'

```

In [9]: # 3.10 Remove rows with invalid codes using variable-specific rules
Code 9 in crossing variable = unknown (self-reported) – excluded
Code 9 in junction_control, road_type, weather, road_surface = un

```
INVALID_CODES = {
    'pedestrian_crossing_physical_facilities_historic': [-1, 9], #
    'junction_detail': [-1, 99],
    'junction_control': [-1, 9],
    'road_type': [-1, 9],
    'speed_limit': [-1, 99],
    'light_conditions': [-1],
    'weather_conditions': [-1, 9],
    'road_surface_conditions': [-1, 9],
    'urban_or_rural_area': [-1],
    'day_of_week': [-1],
}
```

```
df_model = london[FEATURES + [TARGET]].copy()
```

```
for col, codes in INVALID_CODES.items():
    before = len(df_model)
    df_model = df_model[~df_model[col].isin(codes)]
    removed = before - len(df_model)
    if removed > 0:
        print(f"Removed {removed:>5} rows - {col}")

df_model = df_model.dropna()
print(f"\nFinal: {len(df_model):,} rows")
print(f"Severe: {df_model[TARGET].sum():,} ({df_model[TARGET].mean()})")
```

Removed 3690 rows - pedestrian_crossing_physical_facilities_historic

Removed 996 rows - junction_detail

Removed 3718 rows - junction_control

Removed 338 rows - road_type

Removed 396 rows - weather_conditions

Removed 67 rows - road_surface_conditions

Final: 11,762 rows

Severe: 2,266 (19.3%)

In [10]: # 3.11 Verify no remaining missing values

```
missing = df_model[FEATURES].isnull().sum()
print("Missing values:", "None" if missing.sum()==0 else missing[missing>0].index.to_list())
print(f"Shape: {df_model.shape}")
```

Missing values: None

Shape: (11762, 14)

Variable Table

Variable	Source	Type	Description
severe	Collision	Binary	Target: 1 = Fatal or Serious, 0 = Slight
			Pedestrian crossing

pedestrian_crossing_physical_facilities_historic	Collision	Categorical	facility type within 50m
junction_detail	Collision	Categorical	Junction type at collision location
junction_control	Collision	Categorical	Traffic control type at junction
road_type	Collision	Categorical	Road classificatio
speed_limit	Collision	Numeric	Posted speed limit in mph
urban_or_rural_area	Collision	Categorical	Urban or rural classificatio
light_conditions	Collision	Categorical	Lighting at time of collision
weather_conditions	Collision	Categorical	Weather at time of collision
road_surface_conditions	Collision	Categorical	Road surface condition
day_of_week	Collision	Categorical	Day of week
hour	Collision	Numeric	Hour of day (0–23)
peak_hour	Collision	Binary	Peak hour indicator
weekend	Collision	Binary	Weekend indicator

In [11]: *# 3.12 Define categorical and numeric feature groups*

```
CAT_FEATURES = [
    'pedestrian_crossing_physical_facilities_historic',
    'junction_detail',
    'junction_control',
    'road_type',
    'urban_or_rural_area',
    'light_conditions',
    'weather_conditions',
    'road_surface_conditions',
    'day_of_week',
```



```

        'peak_hour',
        'weekend',
    ]

    NUM_FEATURES = [
        'speed_limit',
        'hour',
    ]

```

Exploratory Data Analysis

EDA examined class imbalance, severe rates by crossing facility type and junction context, and the spatial distribution of collisions across London boroughs.

```

In [12]: # 3.13 EDA 1: Severity class distribution

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

severity_counts = df_model['severe'].value_counts().sort_index()
severity_labels = ['Slight (0)', 'Severe (1)']
colors = ['#2B6CB0', '#C81E2A']

axes[0].bar(severity_labels, severity_counts.values,
            color=colors, edgecolor='white', width=0.5)
for bar, count in zip(axes[0].patches, severity_counts.values):
    axes[0].text(bar.get_x() + bar.get_width()/2,
                 bar.get_height() + 50,
                 f'{count:,}', ha='center', va='bottom', fontsize=12)
axes[0].set_title('Collision Count by Severity')
axes[0].set_ylabel('Number of Collisions')
axes[0].set_ylim(0, severity_counts.max() * 1.2)

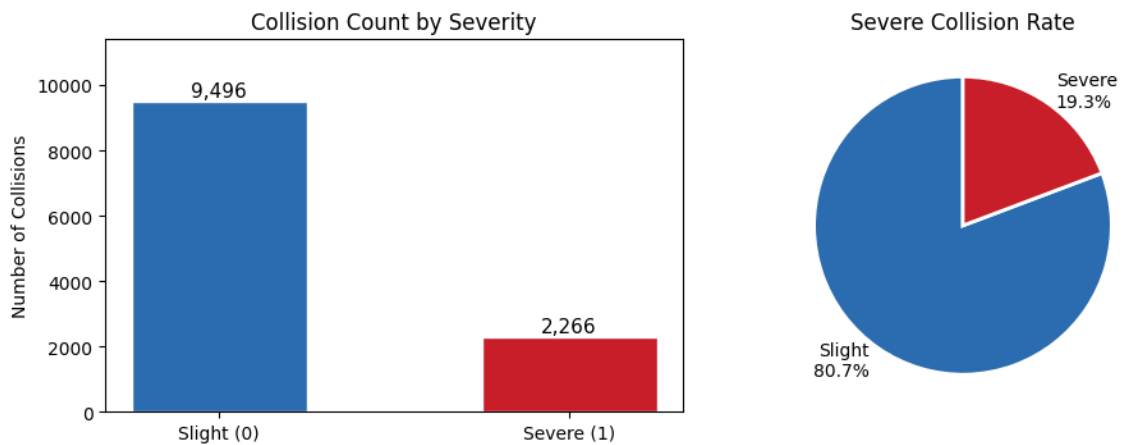
rates = [1 - df_model['severe'].mean(), df_model['severe'].mean()]
axes[1].pie(rates,
            labels=[f'Slight\n{rates[0]*100:.1f}%', f'Severe\n{rates[1]*100:.1f}%'],
            colors=colors,
            startangle=90,
            wedgeprops={'edgecolor': 'white', 'linewidth': 2})
axes[1].set_title('Severe Collision Rate')

plt.suptitle('Class Imbalance: Severe vs Slight Collisions in London')
plt.tight_layout()
plt.show()

print(f"Severe rate: {df_model['severe'].mean():.3f}")
print(f"Imbalance ratio: {(df_model['severe']==0).sum()/df_model['severe'].sum():.3f}")

```

Class Imbalance: Severe vs Slight Collisions in London, 2024



Severe rate: 0.193

Imbalance ratio: 4.2:1

The final dataset contains 11,762 collisions, of which 2,266 (19.3%) are classified as severe (4.2:1 imbalance ratio).

```
In [13]: # 3.14 EDA 2: Severe rate by crossing facility type

facility_labels = {
    0: 'No facility (within 50m)',
    1: 'Zebra crossing',
    4: 'Pelican/Puffin/Toucan crossing',
    5: 'Pedestrian phase at signal junction',
    7: 'Footbridge or subway',
    8: 'Central refuge',
    9: 'Unknown (self-reported)'
}

crossing_rate = (
    df_model.groupby('pedestrian_crossing_physical_facilities_histo
        .agg(severe_rate='mean', count='count')
        .reset_index()
    )
    crossing_rate['label'] = (
        crossing_rate['pedestrian_crossing_physical_facilities_historic
            .map(facility_labels).fillna('Other')
        )
    crossing_rate = crossing_rate.sort_values('severe_rate', ascending=

no_facility_rate = crossing_rate[
    crossing_rate['pedestrian_crossing_physical_facilities_historic
        ]['severe_rate']].values[0]

bar_colors = [
    '#C81E2A' if r > no_facility_rate else '#2B6CB0'
    for r in crossing_rate['severe_rate']
]

fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.barh(crossing_rate['label'],
                crossing_rate['severe_rate'] * 100,
```

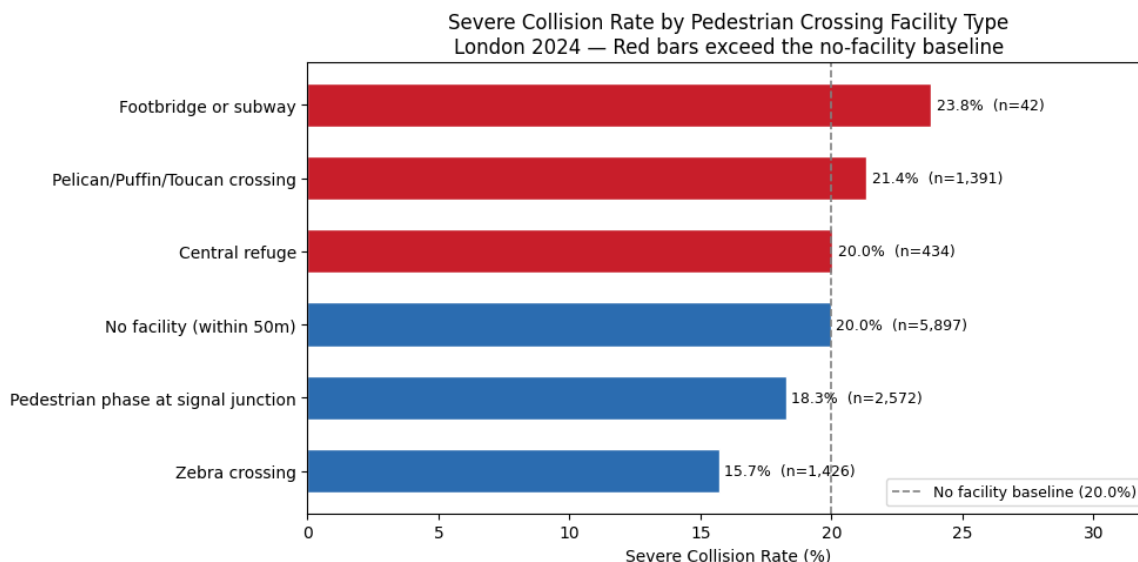
```

        color=bar_colors, edgecolor='white', height=0.6)

for bar, rate, n in zip(bars, crossing_rate['severe_rate'], crossing_rate['n']):
    ax.text(bar.get_width() + 0.2,
            bar.get_y() + bar.get_height()/2,
            f'{rate*100:.1f}% (n={n:,})', va='center', fontsize=9)

ax.axvline(no_facility_rate * 100, color='grey', linestyle='--', label=f'No facility baseline ({no_facility_rate*100:.1f}%')
ax.set_xlabel('Severe Collision Rate (%)')
ax.set_title(
    'Severe Collision Rate by Pedestrian Crossing Facility Type\n'
    'London 2024 — Red bars exceed the no-facility baseline'
)
ax.legend(fontsize=9)
ax.set_xlim(0, crossing_rate['severe_rate'].max() * 100 * 1.35)
plt.tight_layout()
plt.show()

```



Severe rates by crossing facility type

Severe rates vary across facility types. Footbridge or subway recorded the highest rate (23.8%), exceeding the no-facility baseline (20.0%). Zebra crossings recorded the lowest rate (15.7%). These patterns indicate that crossing facility type does not map straightforwardly onto a protective direction.

In [14]: # 3.15 EDA 3: Severe rate by crossing facility × junction type

```

junction_labels = {
    0: 'Not at junction',
    13: 'T or staggered junction',
    16: 'Crossroads',
    17: 'More than 4 arms',
    18: 'Private drive or entrance',
}

crossing_labels_short = {

```

```

0: 'No facility',
1: 'Zebra',
4: 'Pelican/Puffin',
5: 'Signal junction',
7: 'Footbridge/Subway',
8: 'Central refuge',
}

valid_junctions = [0, 13, 16, 17, 18]
plot_df = df_model[df_model['junction_detail'].isin(valid_junctions)]

pivot = (
    plot_df.groupby([
        'pedestrian_crossing_physical_facilities_historic',
        'junction_detail'
    ])['severe'].mean().unstack()
)

count_pivot = (
    plot_df.groupby([
        'pedestrian_crossing_physical_facilities_historic',
        'junction_detail'
    ])['severe'].count().unstack()
)

pivot.index = [crossing_labels_short.get(i, str(i)) for i in pivot.index]
pivot.columns = [junction_labels.get(c, str(c)) for c in pivot.columns]
count_pivot.index = [crossing_labels_short.get(i, str(i)) for i in count_pivot.index]
count_pivot.columns = [junction_labels.get(c, str(c)) for c in count_pivot.columns]

# Annotation: show rate and count together
annot = pivot.copy().astype(object)

for row in pivot.index:
    for col in pivot.columns:
        rate = pivot.loc[row, col]
        n = count_pivot.loc[row, col]

        if pd.notna(rate):
            annot.loc[row, col] = f'{rate:.2f}\n(n={int(n)})'
        else:
            annot.loc[row, col] = 'No data'

# Important:
# seaborn hides NaN cells completely, so create a plotting-only version
# plot_values is used only for rendering; all analysis uses pivot (for calculations)
plot_values = pivot.fillna(-0.01)

# Make values below vmin appear grey
cmap = plt.cm.Reds.copy()
cmap.set_under('#e5e7eb') # light grey for no data

fig, ax = plt.subplots(figsize=(13, 6))

sns.heatmap(
    plot_values,

```

```

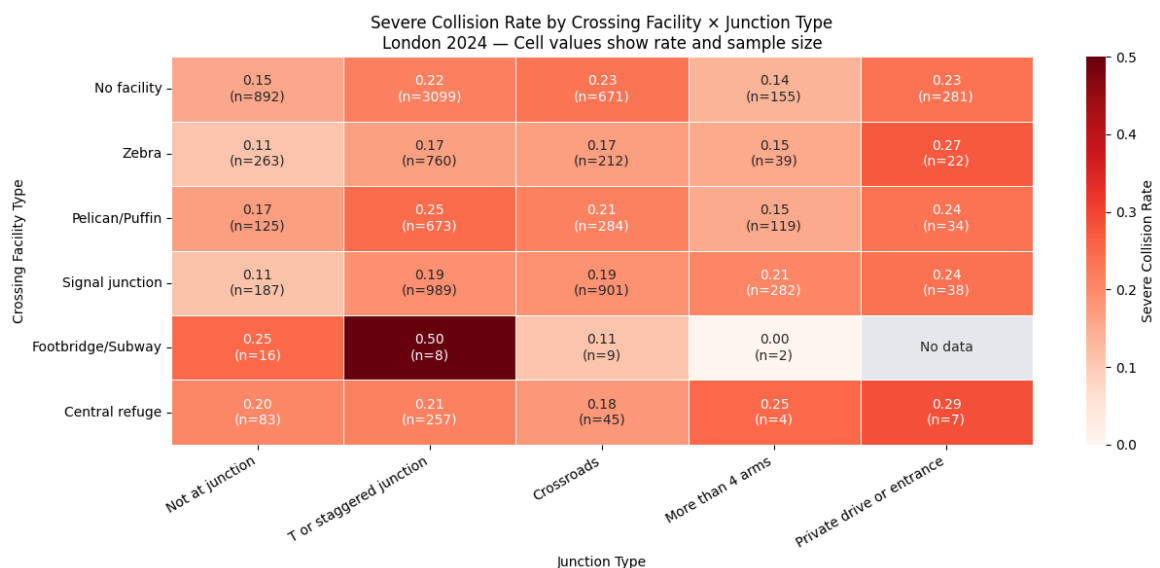
annot=annot,
fmt='',
cmap=cmap,
linewidths=0.5,
linecolor='white',
ax=ax,
vmin=0,
vmax=0.5,
cbar_kws={'label': 'Severe Collision Rate'}
)

ax.set_title(
    'Severe Collision Rate by Crossing Facility x Junction Type\n'
    'London 2024 — Cell values show rate and sample size',
    fontsize=12
)

ax.set_xlabel('Junction Type')
ax.set_ylabel('Crossing Facility Type')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.show()

print(f"Severe rate range: {pivot.min().min():.2f} to {pivot.max().max():.2f}")

```



Severe rate range: 0.00 to 0.50

Crossing facility x junction type interaction

Severe rates across crossing-junction combinations range from 0.00 to 0.50, confirming that junction context is a critical moderating factor. Cells with small sample sizes should be interpreted with caution.

```

In [15]: # 3.16 EDA 4: Spatial distribution of severe collisions in London
# Borough boundary outlines only (no fill), collision points coloured
# Aspect ratio corrected for London latitude; axes removed for clean

import requests
import numpy as np

```

```

fig, ax = plt.subplots(figsize=(11, 10))

# Fetch borough boundary outlines
geojson_url = (
    'https://raw.githubusercontent.com/radoi90/housequest-data/'
    'master/london_boroughs.geojson'
)
response = requests.get(geojson_url, timeout=10)
borough_geojson = response.json()

# Draw borough outlines only – no fill
for feature in borough_geojson['features']:
    geom_type = feature['geometry']['type']
    coords_list = feature['geometry']['coordinates']
    if geom_type == 'Polygon':
        coords_list = [coords_list]
        for polygon_coords in coords_list:
            outer = polygon_coords[0]
            xs = [pt[0] for pt in outer]
            ys = [pt[1] for pt in outer]
            ax.fill(xs, ys, color='#f5f5f5', zorder=0)
            ax.plot(xs, ys, color='#aaaaaa', linewidth=0.7, zorder=1)

map_df = london.loc[df_model.index].copy()

slight = map_df[map_df['severe'] == 0].sample(
    n=min(2000, (map_df['severe'] == 0).sum()),
    random_state=42
)

severe_pts = map_df[map_df['severe'] == 1]

ax.scatter(
    slight['longitude'], slight['latitude'],
    s=3, alpha=0.4, c='#2B6CB0',
    label=f'Slight (sample n={len(slight):,})',
    zorder=2
)

ax.scatter(
    severe_pts['longitude'], severe_pts['latitude'],
    s=6, alpha=0.7, c='#C81E2A',
    label=f'Serious/Fatal (n={len(severe_pts):,})',
    zorder=3
)

# Correct aspect ratio for London latitude (~51.5 degrees)
# 1 degree longitude = cos(51.5°) × 1 degree latitude in physical d.
ax.set_aspect(1 / np.cos(np.radians(51.5)))

ax.set_xlim(-0.55, 0.35)
ax.set_ylim(51.28, 51.72)
ax.axis('off')

ax.set_title(
    'Serious and Fatal Collisions in London, 2024\n'

```

```

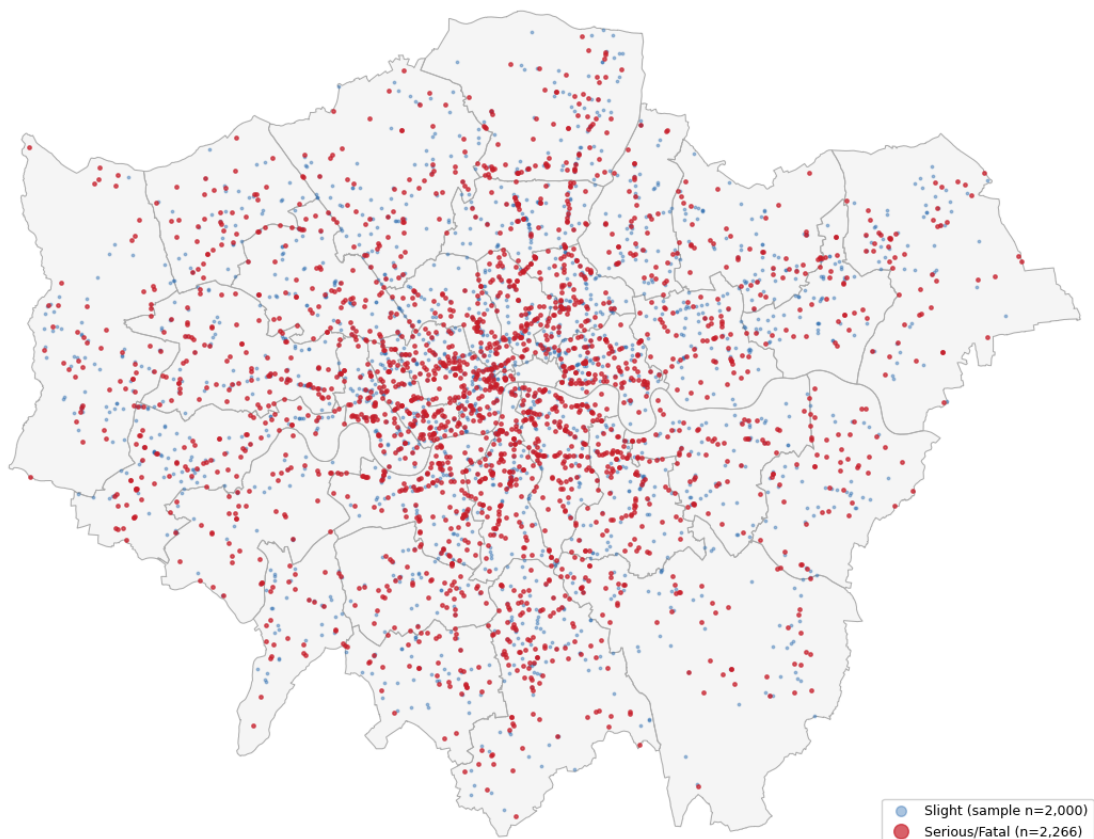
    'Borough boundaries shown; red = serious/fatal, blue = slight (
    fontsize=12, pad=12
)
ax.legend(
    markerscale=4, fontsize=9,
    loc='lower right', framealpha=0.9
)

plt.tight_layout()
plt.show()

print(f"Serious/Fatal plotted: {len(severe_pts):,}")
print(f"Slight sampled:          {len(slight):,}")

```

Serious and Fatal Collisions in London, 2024
Borough boundaries shown; red = serious/fatal, blue = slight (sampled)



Serious/Fatal plotted: 2,266
Slight sampled: 2,000

The spatial map shows that severe collisions are distributed across London rather than concentrated in one single borough. This supports the use of collision-level road-context variables rather than an area-level count model.

```

In [16]: # 3.17 One-hot encode and three-way train/validation/test split
# Train: 60%, Validation: 20%, Test: 20%
# Validation set used for threshold optimisation only
# Test set held out for final evaluation – never used during training

df_encoded = df_model.copy()
df_encoded[CAT_FEATURES] = df_encoded[CAT_FEATURES].astype(str)
df_encoded = pd.get_dummies(df_encoded, columns=CAT_FEATURES)

```

```

X = df_encoded.drop(columns=[TARGET])
y = df_encoded[TARGET]

# Step 1: split off 20% test set
X_trainval, X_test, y_trainval, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Step 2: split remaining 80% into 60% train and 20% validation
# test_size=0.25 of 80% = 20% of total
X_train, X_val, y_train, y_val = train_test_split(
    X_trainval, y_trainval,
    test_size=0.25,
    random_state=42,
    stratify=y_trainval
)

print(f"Total dataset: {len(X):,}")
print(f"Train: {X_train.shape[0]:,} ({X_train.shape[0]/len(X):.3f})")
print(f"Validation: {X_val.shape[0]:,} ({X_val.shape[0]/len(X):.3f})")
print(f"Test: {X_test.shape[0]:,} ({X_test.shape[0]/len(X):.3f})")
print(f"Check total: {X_train.shape[0]+X_val.shape[0]+X_test.shape[0]:,}")
print(f"Features: {X.shape[1]}")
print(f"\nSevere rate:")
print(f"  Train: {y_train.mean():.3f}")
print(f"  Validation: {y_val.mean():.3f}")
print(f"  Test: {y_test.mean():.3f}")

```

```

Total dataset: 11,762
Train: 7,056 (60%)
Validation: 2,353 (20%)
Test: 2,353 (20%)
Check total: 11,762
Features: 53

```

```

Severe rate:
  Train: 0.193
  Validation: 0.193
  Test: 0.193

```

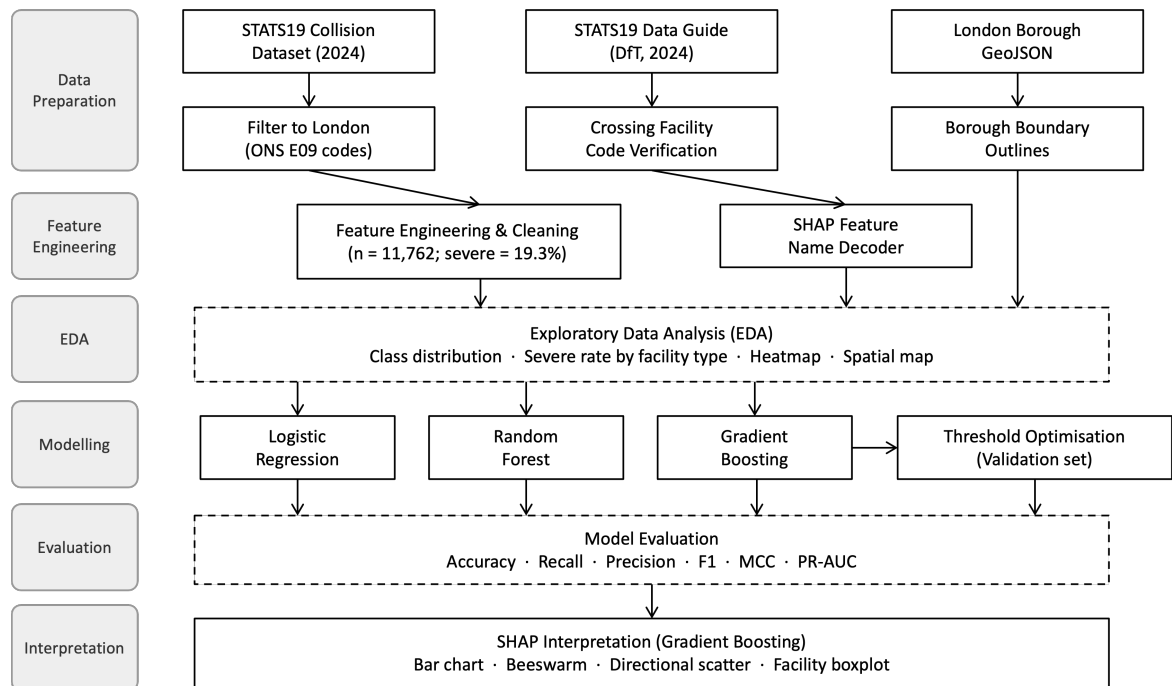
Methodology

[\[go back to the top \]](#)

```

In [17]: from IPython.display import Image, display
display(Image(filename="flowchart.png"))

```

The analysis is conducted at collision level rather than area-level aggregation: the research question concerns the probability of a severe outcome conditional on a collision occurring, not collision frequency across spatial units.

Multicollinearity Assessment

The Variance Inflation Factor (VIF) was computed for numeric features to detect multicollinearity that could distort Logistic Regression coefficients:

$$VIF_i = \frac{1}{1 - R_i^2}$$

Classification Models

Three models are employed, each wrapped in a scikit-learn Pipeline to ensure consistent preprocessing and avoid data leakage.

Logistic Regression serves as a linear baseline. It models the log-odds of a severe collision as a linear combination of features passed through the sigmoid function:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$

Class imbalance is addressed via `class_weight='balanced'`. Features are standardised using `StandardScaler` within the pipeline.

Random Forest constructs an ensemble of T decision trees, each trained on a bootstrap sample, with the final prediction determined by majority vote:

$$\hat{y} = \text{mode}\{h_1(X), h_2(X), \dots, h_T(X)\}$$

Hyperparameters were tuned using GridSearchCV with 5-fold StratifiedKFold cross-validation, optimising average precision (PR-AUC) on the training set only. The selected parameters are reported in the Results section.

Gradient Boosting builds trees sequentially, where each new tree h_m corrects the residuals of the current ensemble:

$$F_m(X) = F_{m-1}(X) + \eta \cdot h_m(X)$$

where η is the learning rate. Hyperparameters were tuned using the same GridSearchCV procedure as Random Forest. The selected parameters are reported in the Results section.

Evaluation and Interpretation

Model performance is evaluated using six metrics. Accuracy is not the primary criterion given class imbalance. Recall is prioritised because a false negative (severe classified as slight) is more consequential for road safety targeting than a false positive.

Recall measures true severe collisions correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision measures predicted severe collisions that are truly severe:

$$\text{Precision} = \frac{TP}{TP + FP}$$

F1-score balances precision and recall:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

MCC provides a balanced summary under class imbalance:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

PR-AUC summarises performance across all thresholds; more informative than ROC-AUC under class imbalance (random baseline = 19.3%).

SHAP was applied to Gradient Boosting to decompose predictions into additive feature contributions. Although Logistic Regression achieved the highest test PR-AUC, Gradient Boosting was selected for SHAP because TreeSHAP efficiently inspects non-linear feature contributions.

Results and discussion

[\[go back to the top \]](#)

```
In [18]: # 5.1 VIF check for numeric features

vif_data = df_model[NUM_FEATURES].copy()

vif_results = pd.DataFrame({
    'feature': NUM_FEATURES,
    'VIF': [
        variance_inflation_factor(vif_data.values, i)
        for i in range(len(NUM_FEATURES))
    ]
}).sort_values('VIF', ascending=False)

print("Variance Inflation Factor:")
print(vif_results.to_string(index=False))
print("\nVIF < 5: low | VIF 5-10: moderate | VIF > 10: problematic")
```

Variance Inflation Factor:

feature	VIF
hour	4.942201
speed_limit	4.942201

VIF < 5: low | VIF 5–10: moderate | VIF > 10: problematic

Both features returned VIF values below 5 (speed_limit: 4.94, hour: 4.94). This does not suggest problematic multicollinearity among the two numeric predictors.

Threshold Optimisation

The default threshold of 0.5 is inappropriate under class imbalance: a naive classifier predicting all collisions as slight would achieve extreme high accuracy. For each model, the threshold was selected on the validation set to minimise deviation from a target accuracy of 0.60–0.65, subject to a minimum recall of 0.20.

```
In [19]: # 5.2 Helper: find model-specific threshold on validation set
# Targets accuracy ~0.6-0.65 with recall >= 0.2
# Threshold selected on validation set and applied once to test set

def find_balanced_threshold(y_true, y_prob, target_accuracy=0.65):
    thresholds = np.arange(0.05, 0.95, 0.01)
    best_thresh = 0.5
    best_diff = 999
    for t in thresholds:
        y_pred = (y_prob >= t).astype(int)
        acc = accuracy_score(y_true, y_pred)
        rec = recall_score(y_true, y_pred, zero_division=0)
        if rec >= 0.2 and abs(acc - target_accuracy) < best_diff:
            best_diff = abs(acc - target_accuracy)
```

```

        best_thresh = t
    return best_thresh

```

In [20]: *# 5.3 Logistic Regression: train on train set, tune threshold on va*
evaluate on test set

```

lr_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('clf', LogisticRegression(
        class_weight='balanced', max_iter=1000, random_state=42
    ))
])
lr_pipeline.fit(X_train, y_train)

y_prob_lr_val = lr_pipeline.predict_proba(X_val)[:, 1]
thresh_lr = find_balanced_threshold(y_val, y_prob_lr_val, target_ac

y_prob_lr = lr_pipeline.predict_proba(X_test)[:, 1]
y_pred_lr = (y_prob_lr >= thresh_lr).astype(int)

print(f"Logistic Regression - threshold: {thresh_lr:.3f}")
print(classification_report(y_test, y_pred_lr,
    target_names=['Slight', 'Severe']))

lr_metrics = {
    'Model': 'Logistic Regression',
    'Threshold': round(thresh_lr, 3),
    'Accuracy': round(accuracy_score(y_test, y_pred_lr), 3),
    'Precision': round(precision_score(y_test, y_pred_lr), 3),
    'Recall': round(recall_score(y_test, y_pred_lr), 3),
    'F1': round(f1_score(y_test, y_pred_lr), 3),
    'MCC': round(matthews_corrcoef(y_test, y_pred_lr), 3),
    'ROC-AUC': round(roc_auc_score(y_test, y_prob_lr), 3),
    'PR-AUC': round(average_precision_score(y_test, y_prob_lr), 3),
}

```

Logistic Regression - threshold: 0.530

	precision	recall	f1-score	support
Slight	0.83	0.65	0.73	1900
Severe	0.22	0.43	0.29	453
accuracy			0.61	2353
macro avg	0.53	0.54	0.51	2353
weighted avg	0.71	0.61	0.64	2353

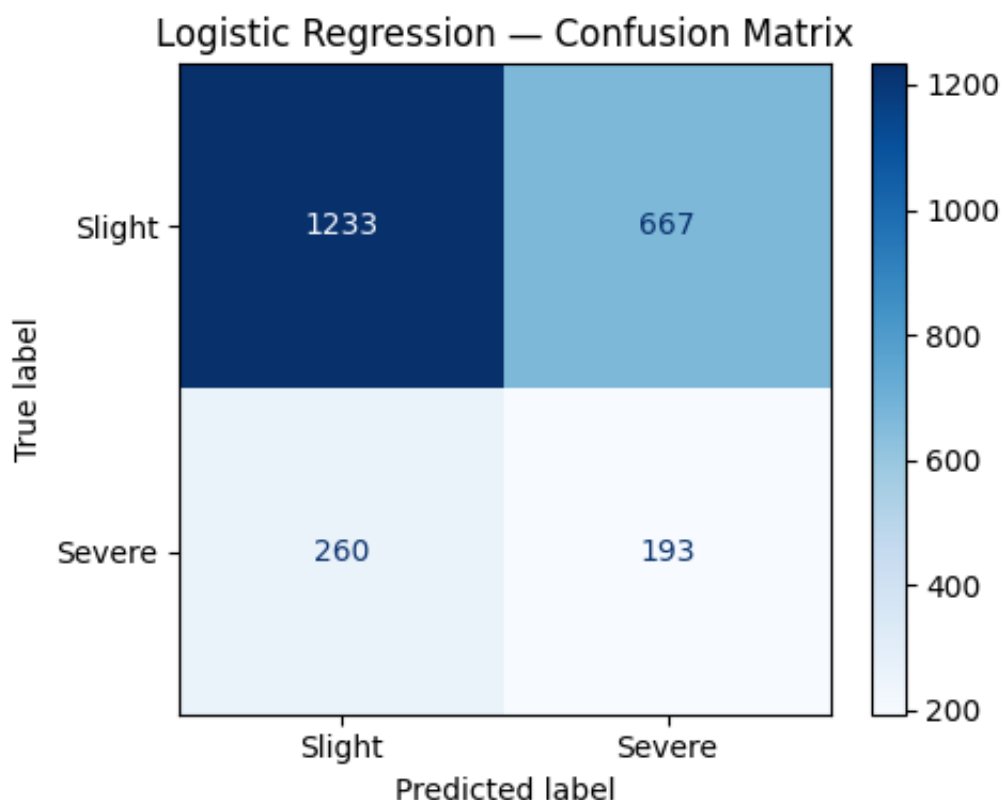
In [21]: *# 5.4a Logistic Regression: confusion matrix*

```

fig, ax = plt.subplots(figsize=(5, 4))
ConfusionMatrixDisplay.from_predictions(
    y_test, y_pred_lr,
    display_labels=['Slight', 'Severe'],
    cmap='Blues', ax=ax
)
ax.set_title('Logistic Regression - Confusion Matrix')

```

```
plt.tight_layout()
plt.show()
```

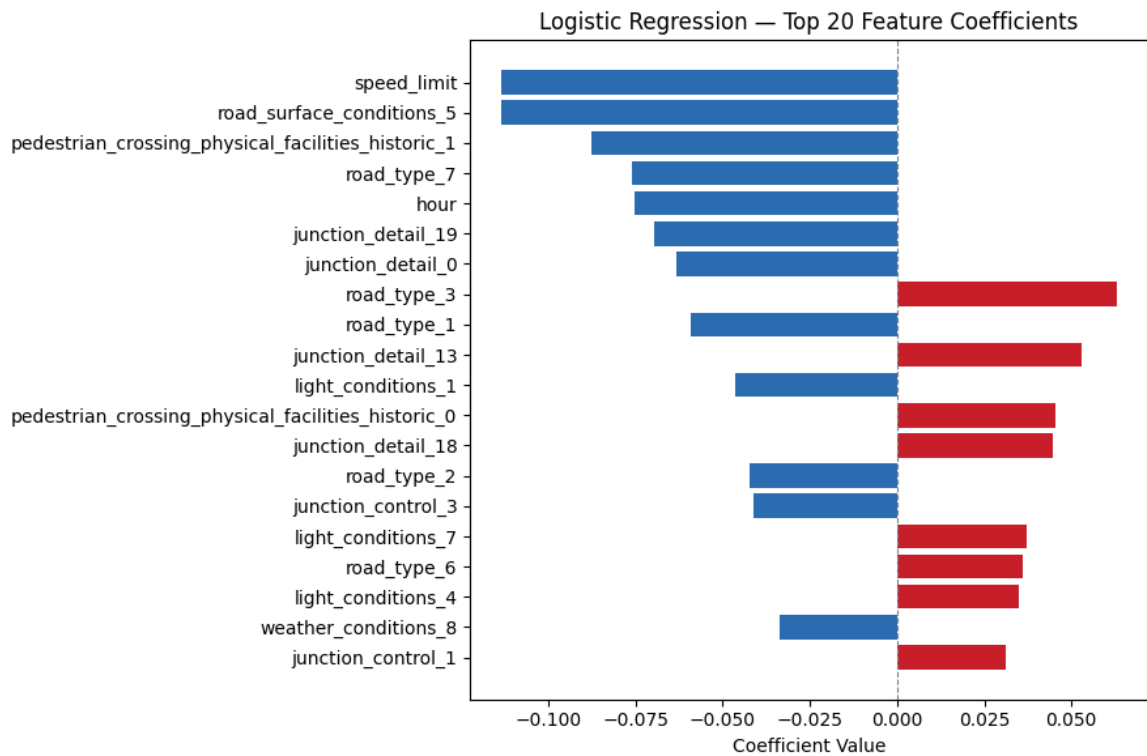


In [22]: *# 5.4b Logistic Regression: top feature coefficients*

```
lr_coef = pd.Series(
    lr_pipeline.named_steps['clf'].coef_[0],
    index=X_train.columns
).sort_values(key=abs, ascending=False).head(20)

colors = ['#C81E2A' if v > 0 else '#2B6CB0' for v in lr_coef.values]

fig, ax = plt.subplots(figsize=(9, 6))
ax.barh(lr_coef.index[::-1], lr_coef.values[::-1], color=colors)
ax.axvline(0, color='grey', linewidth=0.8, linestyle='--')
ax.set_title('Logistic Regression — Top 20 Feature Coefficients')
ax.set_xlabel('Coefficient Value')
plt.tight_layout()
plt.show()
```



```
In [23]: # 5.5a GridSearchCV – Random Forest hyperparameter tuning
# CV only on X_train / y_train; val and test never seen during tuning

from sklearn.model_selection import GridSearchCV, StratifiedKFold

param_grid_rf = {
    'clf__n_estimators': [100, 200],
    'clf__max_depth': [5, 10, None],
    'clf__min_samples_leaf': [2, 5],
    'clf__max_features': ['sqrt', 'log2'],
    'clf__class_weight': ['balanced'],
}

rf_tune_pipe = Pipeline([
    ('passthrough', FunctionTransformer()),
    ('clf', RandomForestClassifier(random_state=42, n_jobs=-1))
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

grid_rf = GridSearchCV(
    rf_tune_pipe,
    param_grid_rf,
    scoring='average_precision', # PR-AUC – appropriate for imbalanced data
    cv=cv,
    n_jobs=-1,
    verbose=1
)

grid_rf.fit(X_train, y_train)

best_rf_params = {
    k.replace('clf__', ''): v
    for k, v in grid_rf.best_params_.items()
}
```

```
print("Best RF parameters:")
print(best_rf_params)
print(f"Best CV PR-AUC: {grid_rf.best_score_:.4f}")
```

Fitting 5 folds for each of 24 candidates, totalling 120 fits

Best RF parameters:

```
{'class_weight': 'balanced', 'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'n_estimators': 200}
```

Best CV PR-AUC: 0.2265

In [24]: # 5.5 Random Forest with Pipeline

```
rf_pipeline = Pipeline([
    ('passthrough', FunctionTransformer()),
    ('clf', RandomForestClassifier(
        **best_rf_params,
        random_state=42, n_jobs=-1
    ))
])
rf_pipeline.fit(X_train, y_train)

y_prob_rf_val = rf_pipeline.predict_proba(X_val)[:, 1]
thresh_rf = find_balanced_threshold(y_val, y_prob_rf_val, target_ac

y_prob_rf = rf_pipeline.predict_proba(X_test)[:, 1]
y_pred_rf = (y_prob_rf >= thresh_rf).astype(int)

print(f"Random Forest - threshold: {thresh_rf:.3f}")
print(classification_report(y_test, y_pred_rf,
    target_names=['Slight', 'Severe']))

rf_metrics = {
    'Model': 'Random Forest',
    'Threshold': round(thresh_rf, 3),
    'Accuracy': round(accuracy_score(y_test, y_pred_rf), 3),
    'Precision': round(precision_score(y_test, y_pred_rf), 3),
    'Recall': round(recall_score(y_test, y_pred_rf), 3),
    'F1': round(f1_score(y_test, y_pred_rf), 3),
    'MCC': round(matthews_corrcoef(y_test, y_pred_rf), 3),
    'ROC-AUC': round(roc_auc_score(y_test, y_prob_rf), 3),
    'PR-AUC': round(average_precision_score(y_test, y_prob_rf), 3),
}
```

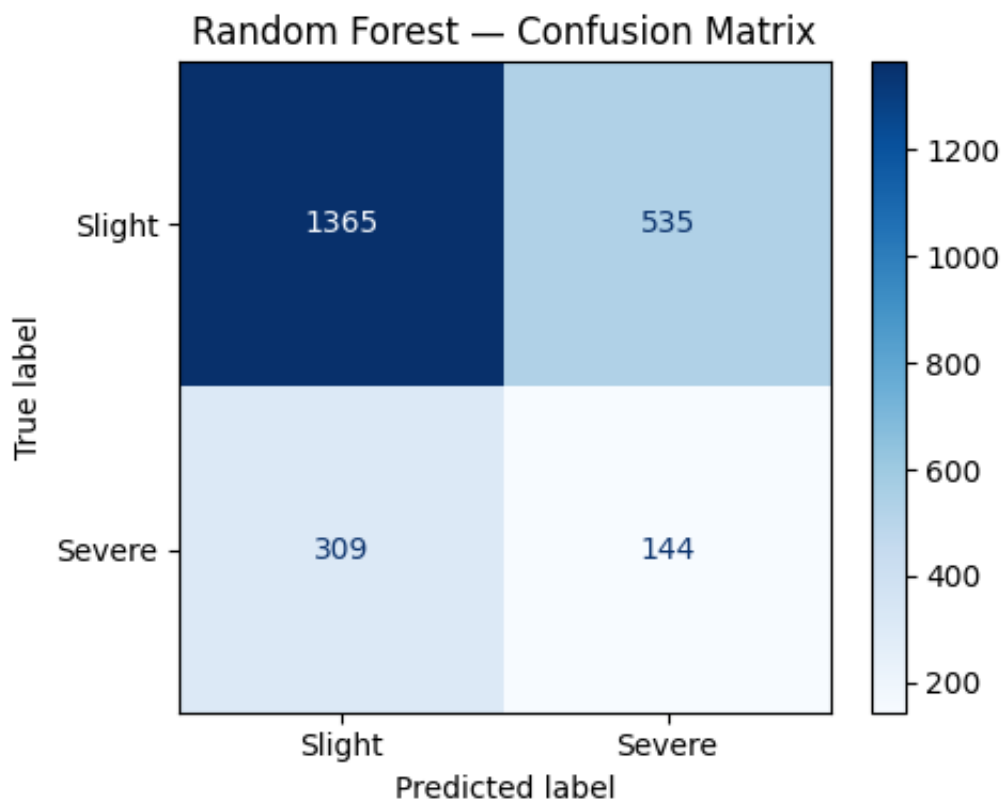
Random Forest - threshold: 0.510

	precision	recall	f1-score	support
Slight	0.82	0.72	0.76	1900
Severe	0.21	0.32	0.25	453
accuracy			0.64	2353
macro avg	0.51	0.52	0.51	2353
weighted avg	0.70	0.64	0.67	2353

In [25]: # 5.6 Random Forest: confusion matrix

```
fig, ax = plt.subplots(figsize=(5, 4))
ConfusionMatrixDisplay.from_predictions(
    y_test, y_pred_rf,
    display_labels=['Slight', 'Severe'],
    cmap='Blues', ax=ax
)
ax.set_title('Random Forest – Confusion Matrix')
plt.tight_layout()
plt.show()

tn, fp, fn, tp = confusion_matrix(y_test, y_pred_rf).ravel()
print(f"True Positives (Severe correctly identified): {tp}")
print(f"False Negatives (Severe missed by model): {fn}")
print(f"False Positives (Slight predicted as Severe): {fp}")
print(f"True Negatives (Slight correctly identified): {tn}")
```



```
True Positives (Severe correctly identified): 144
False Negatives (Severe missed by model): 309
False Positives (Slight predicted as Severe): 535
True Negatives (Slight correctly identified): 1365
```

In [26]: # 5.7a GridSearchCV – Gradient Boosting hyperparameter tuning

```
param_grid_gb = {
    'clf__n_estimators': [100, 200],
    'clf__learning_rate': [0.05, 0.1],
    'clf__max_depth': [3, 4, 5],
    'clf__subsample': [0.8, 1.0],
}

gb_tune_pipe = Pipeline([
    ('passthrough', FunctionTransformer()),
    ('clf', GradientBoostingClassifier(random_state=42))
])
```



```

])

grid_gb = GridSearchCV(
    gb_tune_pipe,
    param_grid_gb,
    scoring='average_precision',
    cv=cv,
    n_jobs=-1,
    verbose=1
)
grid_gb.fit(X_train, y_train)

best_gb_params = {
    k.replace('clf__', ''): v
    for k, v in grid_gb.best_params_.items()
}

print("Best GB parameters:")
print(best_gb_params)
print(f"Best CV PR-AUC: {grid_gb.best_score_:.4f}")

```

Fitting 5 folds for each of 24 candidates, totalling 120 fits

Best GB parameters:

```
{'learning_rate': 0.05, 'max_depth': 3, 'n_estimators': 100, 'subsample': 0.8}
```

Best CV PR-AUC: 0.2201

```

In [27]: # 5.7 Gradient Boosting with Pipeline
gb_pipeline = Pipeline([
    ('passthrough', FunctionTransformer()),
    ('clf', GradientBoostingClassifier(
        **best_gb_params,
        random_state=42
    ))
])
gb_pipeline.fit(X_train, y_train)

y_prob_gb_val = gb_pipeline.predict_proba(X_val)[:, 1]
thresh_gb = find_balanced_threshold(y_val, y_prob_gb_val, target_ac

y_prob_gb = gb_pipeline.predict_proba(X_test)[:, 1]
y_pred_gb = (y_prob_gb >= thresh_gb).astype(int)

print(f"Gradient Boosting - threshold: {thresh_gb:.3f}")
print(classification_report(y_test, y_pred_gb,
    target_names=['Slight', 'Severe']))

gb_metrics = {
    'Model': 'Gradient Boosting',
    'Threshold': round(thresh_gb, 3),
    'Accuracy': round(accuracy_score(y_test, y_pred_gb), 3),
    'Precision': round(precision_score(y_test, y_pred_gb), 3),
    'Recall': round(recall_score(y_test, y_pred_gb), 3),
    'F1': round(f1_score(y_test, y_pred_gb), 3),
    'MCC': round(matthews_corrcoef(y_test, y_pred_gb), 3),
    'ROC-AUC': round(roc_auc_score(y_test, y_prob_gb), 3),
    'PR-AUC': round(average_precision_score(y_test, y_prob_gb), 3),

```

}

Gradient Boosting – threshold: 0.220

	precision	recall	f1-score	support
Slight	0.82	0.70	0.76	1900
Severe	0.22	0.34	0.26	453
accuracy			0.63	2353
macro avg	0.52	0.52	0.51	2353
weighted avg	0.70	0.63	0.66	2353

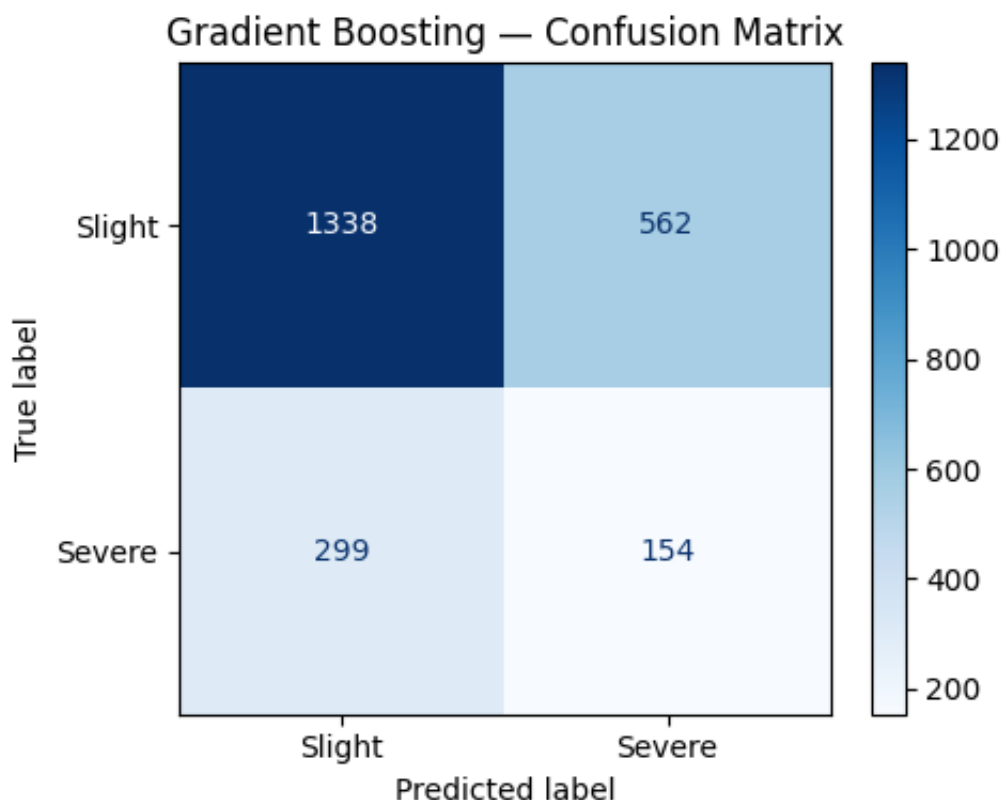
In [28]: # 5.8 Gradient Boosting: confusion matrix

```

fig, ax = plt.subplots(figsize=(5, 4))
ConfusionMatrixDisplay.from_predictions(
    y_test, y_pred_gb,
    display_labels=['Slight', 'Severe'],
    cmap='Blues', ax=ax
)
ax.set_title('Gradient Boosting – Confusion Matrix')
plt.tight_layout()
plt.show()

tn, fp, fn, tp = confusion_matrix(y_test, y_pred_gb).ravel()
print(f"True Positives (Severe correctly identified): {tp}")
print(f"False Negatives (Severe missed by model): {fn}")
print(f"False Positives (Slight predicted as Severe): {fp}")
print(f"True Negatives (Slight correctly identified): {tn}")

```



True Positives (Severe correctly identified): 154
 False Negatives (Severe missed by model): 299
 False Positives (Slight predicted as Severe): 562
 True Negatives (Slight correctly identified): 1338

```
In [29]: # 5.9 Full model comparison table and PR curves

comparison = pd.DataFrame([lr_metrics, rf_metrics, gb_metrics])
cols = ['Model', 'Threshold', 'Accuracy', 'Precision', 'Recall', 'F1', 'MCC', 'ROC']

fig, ax = plt.subplots(figsize=(11, 2))
ax.axis('off')

table = ax.table(
    cellText=comparison[cols].values.tolist(),
    colLabels=cols,
    cellLoc='center',
    loc='center'
)
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1, 2.2)

# Header: dark grey
for j in range(len(cols)):
    table[0, j].set_facecolor('#374151')
    table[0, j].set_text_props(color='white', fontweight='bold')

# Rows: alternating white and light grey
row_bg = ['#ffffff', '#f3f4f6', '#ffffff']
for i in range(1, 4):
    for j in range(len(cols)):
        table[i, j].set_facecolor(row_bg[i-1])

# Highlight best value per metric in soft amber
metric_cols = ['Accuracy', 'Precision', 'Recall', 'F1', 'MCC', 'ROC']
for col_name in metric_cols:
    col_idx = cols.index(col_name)
    values = comparison[col_name].values
    best_row = values.argmax() + 1
    table[best_row, col_idx].set_facecolor('#fde68a')
    table[best_row, col_idx].set_text_props(fontweight='bold')

ax.set_title(
    'Model Performance Comparison - Optimal Thresholds\n'
    'Bold = best value per metric',
    fontsize=12, pad=20
)

plt.tight_layout()
plt.show()

print("=== Model Comparison ===")
print(comparison[cols].to_string(index=False))

# PR curves
fig, ax = plt.subplots(figsize=(8, 6))
```

```

for name, y_prob, color in [
    ('Logistic Regression', y_prob_lr, '#2B6CB0'),
    ('Random Forest', y_prob_rf, '#D97706'),
    ('Gradient Boosting', y_prob_gb, '#276749'),
]:
    prec, rec, _ = precision_recall_curve(y_test, y_prob)
    ap = average_precision_score(y_test, y_prob)
    ax.plot(rec, prec, label=f'{name} (PR-AUC={ap:.3f})', linewidth=1)

ax.axhline(y_test.mean(), color='grey', linestyle='--', linewidth=1,
            label=f'Random baseline ({y_test.mean():.3f})')
ax.set_xlabel('Recall')
ax.set_ylabel('Precision')
ax.set_title('Precision-Recall Curves – All Models\nLondon Serious')
ax.legend(fontsize=9)
ax.set_xlim(0, 1)
ax.set_ylim(0, 1)
plt.tight_layout()
plt.show()

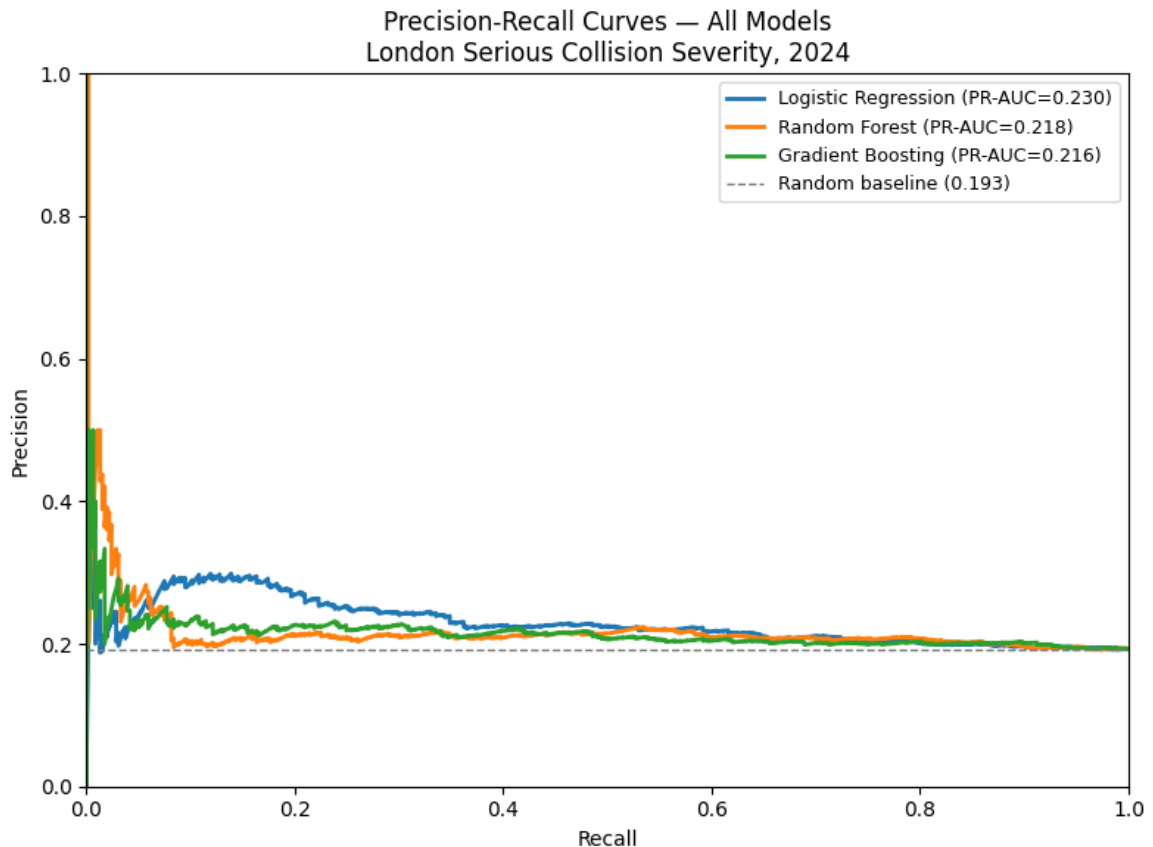
```

Model Performance Comparison — Optimal Thresholds
Bold = best value per metric

Model	Threshold	Accuracy	Precision	Recall	F1	MCC	ROC-AUC	PR-AUC
Logistic Regression	0.53	0.606	0.224	0.426	0.294	0.061	0.557	0.23
Random Forest	0.51	0.641	0.212	0.318	0.254	0.032	0.541	0.218
Gradient Boosting	0.22	0.634	0.215	0.34	0.263	0.038	0.537	0.216

=== Model Comparison ===

	Model	Threshold	Accuracy	Precision	Recall	F1
MCC	ROC-AUC	PR-AUC				
061	0.557	0.230	0.53	0.606	0.224	0.426 0.294 0.
032	0.541	0.218	0.51	0.641	0.212	0.318 0.254 0.
038	0.537	0.216	0.22	0.634	0.215	0.340 0.263 0.



Model comparison

All three models achieved PR-AUC above the random baseline of 0.193. Logistic Regression achieved the highest PR-AUC (0.230) and recall (0.426) with a threshold of 0.530, suggesting the linear baseline captured slightly more predictive signal than the tuned tree-based models. Random Forest achieved the highest accuracy (0.641) but the lowest recall (0.318). Gradient Boosting produced a recall of 0.340 and PR-AUC of 0.216 with a threshold of 0.220. MCC values are low across all models (0.032–0.061), confirming modest gains above the no-information baseline. This reflects the observability constraints of STATS19, which does not record impact speed or driver behaviour.

```
In [30]: # 5.10 SHAP
import shap
X_shap = X_test.copy()
gb_clf = gb_pipeline.named_steps['clf']
explainer = shap.TreeExplainer(gb_clf)
shap_values = explainer.shap_values(X_shap)
```

```
In [31]: # 5.10b Read STATS19 data guide from GitHub to verify crossing faci
# then build feature name decoder for SHAP plots

import io
import requests

DATA_GUIDE_URL = (
    'https://raw.githubusercontent.com/Alexia418/'
```

```

'DSSS_Assessment/main/'
'dft-road-casualty-statistics-road-safety-open-dataset-data-guide'
)

response = requests.get(DATA_GUIDE_URL)
guide_raw = pd.read_excel(
    io.BytesIO(response.content),
    sheet_name='2024_code_list'
)

# Extract and display crossing facility codes for verification
crossing_codes = guide_raw[
    guide_raw['field name'] == 'pedestrian_crossing_physical_facilities'
][['field name', 'code/format', 'label']].dropna(subset=['label'])

print("Crossing facility codes (from DfT data guide 2024):")
print(crossing_codes.to_string(index=False))

# Build feature name decoder using verified STATS19 coding
feature_name_map = {
    'speed_limit': 'Speed limit (mph)',
    'hour': 'Hour of day',

    'pedestrian_crossing_physical_facilities_historic_0': 'Crossing',
    'pedestrian_crossing_physical_facilities_historic_1': 'Crossing',
    'pedestrian_crossing_physical_facilities_historic_4': 'Crossing',
    'pedestrian_crossing_physical_facilities_historic_5': 'Crossing',
    'pedestrian_crossing_physical_facilities_historic_7': 'Crossing',
    'pedestrian_crossing_physical_facilities_historic_8': 'Crossing',

    'junction_detail_0': 'Junction: Not at junction',
    'junction_detail_13': 'Junction: T or staggered',
    'junction_detail_16': 'Junction: Crossroads',
    'junction_detail_17': 'Junction: More than 4 arms',
    'junction_detail_18': 'Junction: Private drive',
    'junction_detail_19': 'Junction: Mini-roundabout',

    'junction_control_0': 'Control: Not at junction',
    'junction_control_1': 'Control: Authorised person',
    'junction_control_2': 'Control: Auto traffic signal',
    'junction_control_3': 'Control: Stop sign',
    'junction_control_4': 'Control: Give way/uncontrolled',

    'road_type_1': 'Road: Roundabout',
    'road_type_2': 'Road: One-way street',
    'road_type_3': 'Road: Dual carriageway',
    'road_type_6': 'Road: Single carriageway',
    'road_type_7': 'Road: Slip road',
    'road_type_12': 'Road: One-way/Slip road',

    'light_conditions_1': 'Light: Daylight',
    'light_conditions_4': 'Light: Darkness (street lit)',
    'light_conditions_5': 'Light: Darkness (unlit)',
    'light_conditions_6': 'Light: Darkness (no lighting)',
    'light_conditions_7': 'Light: Darkness (lighting unknown)',

```

```

'weather_conditions_1': 'Weather: Fine',
'weather_conditions_2': 'Weather: Raining',
'weather_conditions_3': 'Weather: Snowing',
'weather_conditions_4': 'Weather: Fine + wind',
'weather_conditions_5': 'Weather: Raining + wind',
'weather_conditions_6': 'Weather: Snowing + wind',
'weather_conditions_7': 'Weather: Fog or mist',
'weather_conditions_8': 'Weather: Other',

'road_surface_conditions_1': 'Surface: Dry',
'road_surface_conditions_2': 'Surface: Wet or damp',
'road_surface_conditions_3': 'Surface: Snow',
'road_surface_conditions_4': 'Surface: Frost or ice',
'road_surface_conditions_5': 'Surface: Flood',

'urban_or_rural_area_1': 'Area: Urban',
'urban_or_rural_area_2': 'Area: Rural',

'day_of_week_1': 'Day: Sunday',
'day_of_week_2': 'Day: Monday',
'day_of_week_3': 'Day: Tuesday',
'day_of_week_4': 'Day: Wednesday',
'day_of_week_5': 'Day: Thursday',
'day_of_week_6': 'Day: Friday',
'day_of_week_7': 'Day: Saturday',

'peak_hour_0': 'Off-peak hour',
'peak_hour_1': 'Peak hour (07-09 / 16-19)',
'weekend_0': 'Weekday',
'weekend_1': 'Weekend',
}

X_shap_named = X_shap.rename(columns=feature_name_map)

decoded = sum(1 for c in X_shap.columns if c in feature_name_map)
print(f"\nDecoder built: {len(feature_name_map)} entries")
print(f"Columns decoded: {decoded} of {len(X_shap.columns)}")

```

Crossing facility codes (from DfT data guide 2024):

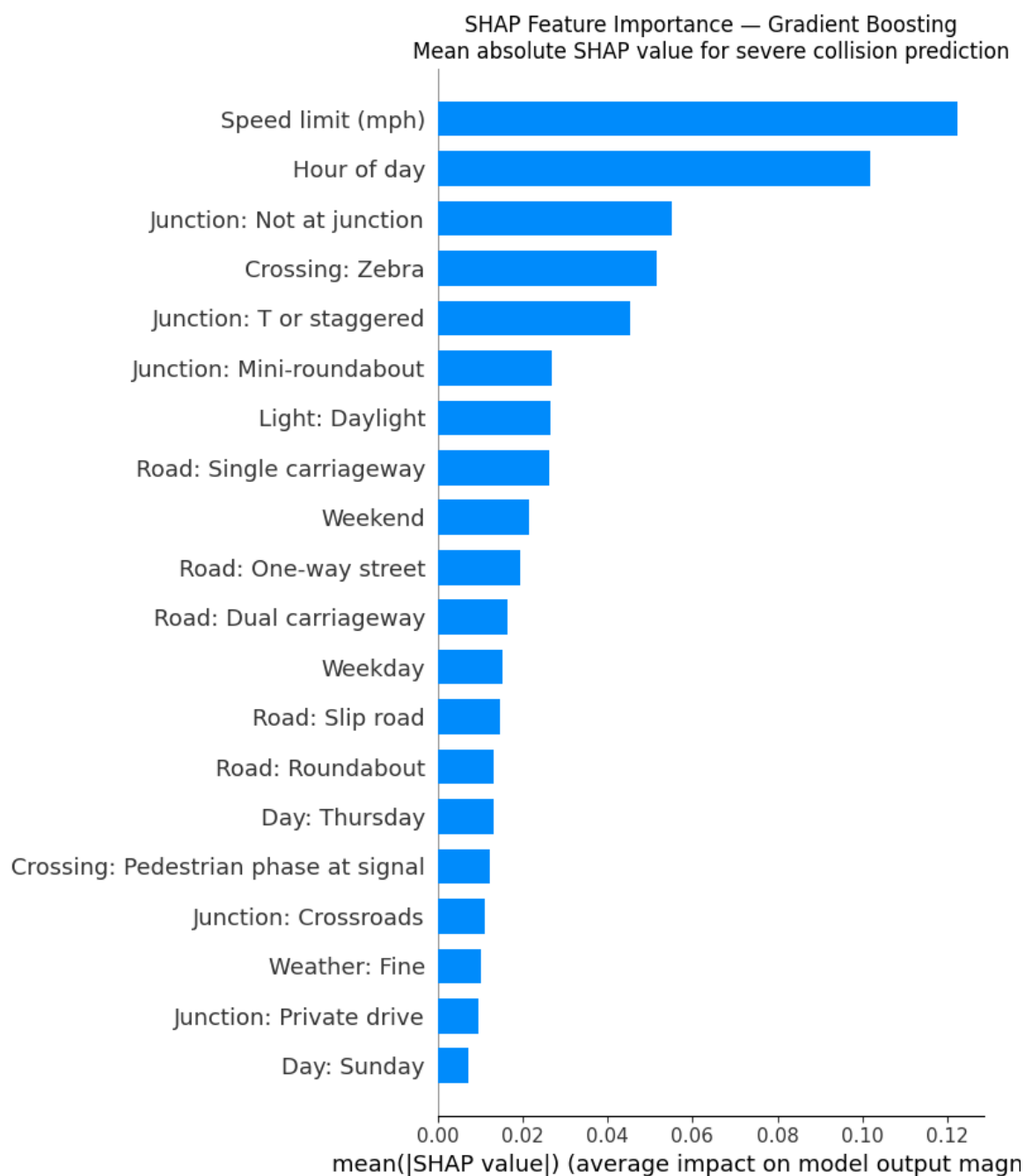
	field name	code/format
label		
pedestrian_crossing_physical_facilities_historic		0
No physical crossing facilities within 50 metres		
pedestrian_crossing_physical_facilities_historic		1
Zebra		
pedestrian_crossing_physical_facilities_historic		4
n, puffin, toucan or similar non-junction pedestrian light crossing		
pedestrian_crossing_physical_facilities_historic		5
Pedestrian phase at traffic signal junction		
pedestrian_crossing_physical_facilities_historic		7
Footbridge or subway		
pedestrian_crossing_physical_facilities_historic		8
Central refuge		
pedestrian_crossing_physical_facilities_historic		-1
Data missing or out of range		
pedestrian_crossing_physical_facilities_historic		9
unknown (self reported)		

Decoder built: 56 entries

Columns decoded: 53 of 53

In [32]: *# 5.11 SHAP bar chart with readable feature names*

```
shap.summary_plot(
    shap_values, X_shap_named,
    plot_type='bar', max_display=20, show=False
)
plt.title(
    'SHAP Feature Importance – Gradient Boosting\n'
    'Mean absolute SHAP value for severe collision prediction',
    fontsize=12
)
plt.tight_layout()
plt.show()
```

In [33]: *# 5.12 SHAP beeswarm with readable feature names*

```
shap.summary_plot(
    shap_values, X_shap_named,
    plot_type='dot', max_display=20, show=False
)
plt.title(
    'SHAP Summary Plot — Gradient Boosting\n'
    'Direction and magnitude of feature contributions',
    fontsize=12
)
plt.tight_layout()
plt.show()
```



In [34]: *# 5.12b SHAP directional scatter – speed_limit, hour, and crossing*
Three panels: continuous predictors (scatter+lowess) + categorical
Together these directly answer RQ2: direction and magnitude of key

```
import statsmodels.api as sm

fig, axes = plt.subplots(1, 3, figsize=(19, 5))

# --- Panel 1 & 2: continuous features (scatter + lowess) ---
focus_features = {
    'speed_limit': 'Speed Limit (mph)',
    'hour': 'Hour of Day (0-23)',
}

for ax, (feat, label) in zip(axes[:2], focus_features.items()):
    if feat not in X_shap.columns:
        continue
```

```

feat_idx    = list(X_shap.columns).index(feat)
feat_vals   = X_shap[feat].values
shap_vals   = shap_values[:, feat_idx]

sort_idx    = np.argsort(feat_vals)
x_sorted    = feat_vals[sort_idx]
shap_sorted = shap_vals[sort_idx]

lowess_result = sm.nonparametric.lowess(
    shap_sorted, x_sorted, frac=0.3, return_sorted=True
)

ax.scatter(feat_vals, shap_vals,
           alpha=0.25, s=15, color='#2B6CB0', zorder=2)
ax.plot(lowess_result[:, 0], lowess_result[:, 1],
        color='#C81E2A', linewidth=2.5, zorder=3, label='Lowess')
ax.axhline(0, color='grey', linewidth=0.8, linestyle='--', alpha=0.5)

ax.set_xlabel(label, fontsize=11)
ax.set_ylabel('SHAP value\n(positive = higher severe probability)')
ax.set_title(f'SHAP Directional Pattern\n{label}', fontsize=12)
ax.legend(fontsize=9)
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# --- Panel 3: crossing facility type boxplot ---
ax3 = axes[2]

# Crossing facility labels (matching one-hot column names after encoding)
crossing_display = {
    'pedestrian_crossing_physical_facilities_historic_0': 'No facilities',
    'pedestrian_crossing_physical_facilities_historic_1': 'Zebra',
    'pedestrian_crossing_physical_facilities_historic_4': 'Pelican/Zebr',
    'pedestrian_crossing_physical_facilities_historic_5': 'Signal pedestrian',
    'pedestrian_crossing_physical_facilities_historic_7': 'Footbridge',
    'pedestrian_crossing_physical_facilities_historic_8': 'Central island'
}

box_data    = []
box_labels  = []
box_colors  = []

for col, display_name in crossing_display.items():
    if col not in X_shap.columns:
        continue
    col_idx    = list(X_shap.columns).index(col)
    shap_vals  = shap_values[:, col_idx]
    # Only include rows where this facility type was present (one-hot encoding)
    mask       = X_shap[col].values == 1
    if mask.sum() < 5:
        continue
    box_data.append(shap_vals[mask])
    box_labels.append(display_name)
    # Red if median SHAP > 0 (risk signal), blue if < 0 (protective signal)
    box_colors.append('#C81E2A' if np.median(shap_vals[mask]) > 0 else '#1F77B4')

```

SHAP Directional Analysis — Gradient Boosting
Continuous predictors (panels 1-2) and crossing facility context (panel 3)

SHAP Directional Pattern
Speed Limit (mph)

SHAP value (positive = higher severe probability)

Lowest trend

Speed Limit (mph)

SHAP Directional Pattern
Hour of Day (0-23)

SHAP value (positive = higher severe probability)

Lowest trend

Hour of Day (0-23)

SHAP by Crossing Facility Type
(red = risk signal, blue = protective)

SHAP value (positive = higher severe probability)

No effect (SHAP = 0)

No facility, Zebra, Pelican/Puffin, Signal phase, Roundabout/subway, Central refuge

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non-linear pattern across the day.

In [35]: *# 5.13 SHAP values for crossing facility features specifically*

```
crossing_cols = [
    c for c in X_shap_named.columns
    if c.startswith('Crossing:')
]

crossing_shap = pd.Series({
    col: np.abs(
        shap_values[:, list(X_shap_named.columns).index(col)]
    ).mean()
    for col in crossing_cols
}).sort_values(ascending=False)

print("Mean absolute SHAP – crossing facility categories:")
print(crossing_shap.to_string())
```

Mean absolute SHAP – crossing facility categories:

Crossing: Zebra	0.051520
Crossing: Pedestrian phase at signal	0.012277
Crossing: No facility (within 50m)	0.006695
Crossing: Central refuge	0.004931
Crossing: Pelican/Puffin/Toucan	0.003355
Crossing: Footbridge or subway	0.000893

Crossing facility SHAP values

Among crossing facility categories, Zebra crossing recorded the highest mean absolute SHAP value (0.0515), followed by Pedestrian phase at signal junction (0.0123), No facility within 50m (0.0067), Central refuge (0.0049), Pelican/Puffin/Toucan (0.0034), and Footbridge or subway (0.0009). All crossing facility SHAP values are substantially lower than those for speed_limit and hour. The SHAP boxplot shows the directional contribution of each facility type: where the median SHAP for a facility type is negative, the model associates its presence with a lower predicted probability of a severe outcome relative to the baseline. These results indicate that crossing facility type is not a strong independent predictor of severity in this model, and that speed limit and time of day dominate the prediction.

Limitations

The analysis is conditional on a collision being recorded in STATS19: slight collisions may be under-reported, affecting class distribution. Pedestrian exposure data is absent, crossing facilities are typically installed at high-exposure locations, so lower severe rates at facility locations may reflect higher footfall rather than a protective effect. All model outputs represent predictive associations, not causal effects. Spatial autocorrelation between nearby collisions is not modelled.

Conclusion

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This study examined whether pedestrian crossing facilities in London are associated with a lower probability of serious or fatal collision outcomes, using STATS19 2024 records.

Addressing RQ1, severe rates vary across facility types (23.8% for Footbridge; 15.7% for Zebra) and across crossing-junction combinations (0.00–0.50), confirming junction context as a critical moderating factor.

Addressing RQ2, all models exceeded the random PR-AUC baseline of 0.193. Logistic Regression performed best (PR-AUC 0.230, recall 0.426). SHAP identified speed limit and hour as dominant predictors. All crossing facility SHAP values are substantially lower, indicating facility type is not a strong independent predictor. Its effect depends on junction context and road speed.

Future research should incorporate pedestrian flow data to separate infrastructure effects from exposure effects.

References

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